AML Machine Learning & Analytics Project — Technical and Audit Documentation

Document purpose: Record the business rationale, notebook data preparation, exploratory statistical tests, interpretations, and implementation considerations for the AML analytics prototype.

Evidence basis: Review of `AML.ipynb` and its embedded outputs. The notebook reads `Medium_Trans.csv` (31,251,483 rows × 11 columns), creates a resampled and imputed 150,000-row analysis dataset, and separately uses timestamps from the original file for daily-volume analysis. This report documents the implemented workflow; it does not represent a validated production AML control or regulatory approval.

1. EXECUTIVE SUMMARY & BUSINESS CONTEXT

Problem statement

Financial institutions process large volumes of payments across channels, currencies, counterparties, and account networks. Static threshold and rule-based monitoring can be difficult to maintain as products, customer behavior, and laundering typologies evolve. Broad rules can generate high alert volumes and false positives, consuming investigator capacity; narrow rules may fail to identify less obvious, multi-step, or rapidly changing patterns. Manual review alone is not scalable and can delay detection and escalation.

Business objective

The project explores a shift toward an intelligent, data-driven monitoring capability that can prioritize potentially high-risk accounts and transactions for trained compliance personnel. Statistical analysis identifies candidate signals and data characteristics; these findings motivate a subsequent supervised machine-learning classification model. Model output should support—not replace—risk-based human review, investigation, escalation, and reporting decisions.

Regulatory importance

An effective AML and counter-terrorist financing program supports applicable obligations under the Bank Secrecy Act (BSA), its implementing rules, sanctions and terrorist-financing controls, and relevant local requirements. Internationally, program design should be consistent with applicable Financial Action Task Force (FATF) risk-based principles. Analytics can strengthen detection and prioritization, but the institution remains responsible for governance, recordkeeping, suspicious activity escalation/reporting, independent testing, and demonstrable control effectiveness. This exploratory notebook is not legal advice and does not establish compliance by itself.

2. DATA ENGINEERING & PRE-PROCESSING ARCHITECTURE

2.1 Source and baseline resampling

The source file reported by the notebook contains **31,251,483 records and 11 fields**. The code constructs exactly **150,000 rows** with a nominal **70/30 target distribution**:

Class	Label	Target count	Sampling approach
Legitimate	0	105,000 (70%)	Random downsampling without replacement (`random_state=42`)
Laundering-labelled	1	45,000 (30%)	Random oversampling with replacement (`random_state=42`)
Total		**150,000**	Concatenated and shuffled (`random_state=42`)

Let the available class subsets be D_0 and D_1. The constructed set is a sample of 105,000 distinct records from D_0 combined with 45,000 draws from D_1 with replacement. This addresses computational scale and makes the minority class more visible in exploratory comparisons. A 30% positive share is an engineered analytical prevalence, not an estimate of real-world laundering prevalence.

Interpretation and controls: Oversampling with replacement duplicates some positive-class source records. Observations are therefore not all independent; naïve inferential standard errors or random train/test splits can be optimistic if copies of a source transaction cross partitions. The engineered class prior also changes probability calibration, precision, alert volume, and decision thresholds relative to deployment. For model development, preserve a source-record identifier, split by original entity/time before resampling, fit resampling only on training data, retain a representative untouched test set, and calibrate against operational prevalence. Version the sampling approach, source snapshot, seed, and class counts.

2.2 Missing-value injection simulation

The notebook uses `np.random.default_rng(42)` and, for each column, draws an independent target fraction $r_j \sim \text{Uniform}(0.005, 0.025)$. It selects approximately $n_j = \text{floor}(N \times r_j)$ eligible rows for that column, without replacement, where eligibility requires that the row currently has fewer than three injected missing values. A row-level counter caps injected missingness at **three fields per row**. The final dataset remains 150,000 × 11.

This deliberate perturbation is a **simulation**, not missingness observed in the source. It can test whether a basic cleaning path executes under sparse, dispersed data-quality defects. Actual missing telemetry may arise from source-system outages, interface failures, product-specific collection gaps, or process changes; it may be correlated with channel, institution, time, or risk. Random injection does not reproduce those mechanisms (MCAR/MAR/MNAR), nor establish production data quality. The fixed seed supports repeatability; reproducibility also requires stable source data and software versions.

The notebook's displayed injected-null counts range from approximately 1,213 to 3,071 by column, and no row received more than three injected nulls. Missingness is also injected into `Is Laundering`, which requires particular care because target labels should not ordinarily be synthetically altered or imputed.

2.3 Imputation methodology

For each field with missing values, the notebook applies a dtype-based fallback:

- **Numeric fields:** replace missing values with the observed column median, $\tilde{x}_j = \text{median}(x_j)$. The median is less sensitive than the mean to extreme transaction values and heavy right tails.
- **Non-numeric fields:** replace missing values with the most frequent non-null category, $\text{mode}(x_j)$. If no mode exists, the code leaves missingness unresolved; the reported run showed zero nulls after filling.

This is a simple baseline, not a complete production imputation policy. For the target, median/mode replacement is especially problematic: it assigns an assumed class to records whose label was artificially removed and can change the actual post-cleaning class distribution. Target rows with unknown labels should instead be excluded from supervised training or restored from authoritative labels; the target must never be imputed as a predictor. For modeling, estimate imputation parameters on training folds only to prevent leakage, consider explicit missingness indicators, and assess whether missingness itself carries risk or operational meaning. Preserve raw values and an auditable transformation log.

2.4 Resulting analytical dataset

The cleaned file is saved as `LI-Medium_Trans_150k_70-30_cleaned.csv`. The class proportions printed immediately after resampling are 70%/30%; because missing target values are subsequently imputed, those proportions should not be assumed to remain exact in the final file without recounting the target after imputation. The transaction amount test filters to **US Dollar receiving-currency records**. The payment-format association and amount-by-format analyses then use this filtered `df1` subset, not necessarily all 150,000 rows. The systems-volume analysis instead reads timestamps from the original full source file.

3. TRANSACTIONS METADATA CHEATSHEET

The notebook uses the following 11-column schema. Types below reflect observed/expected representation in the notebook and sample rows; formal source-system data contracts were not provided.

Field	Expected structure / analytical role	Notes
`Timestamp`	Date-time string, parsed as timestamp for daily aggregation	Sample format `YYYY/MM/DD HH:MM`; source timezone and event-time semantics are unspecified.
`From Bank`	Numeric-looking institution identifier	Identifier, not a continuous measure; preserve as categorical/string in production.
`Account`	Originating account identifier	Alphanumeric ID; sensitive/linkable entity data requiring access controls.
`To Bank`	Numeric-looking beneficiary institution identifier	Identifier; numeric dtype does not imply ordinal meaning.
`Account.1`	Beneficiary account identifier	Alphanumeric ID; preserve leading characters and formatting.
`Amount Received`	Numeric transaction amount received	Currency is specified separately by `Receiving Currency`; values across currencies are not directly comparable without normalization.
`Receiving Currency`	Categorical currency name/ code	Examples include US Dollar, Euro, Australian Dollar. No FX conversion is shown.
`Amount Paid`	Numeric amount paid	Currency specified separately by `Payment Currency`; may differ from received amount/currency.
`Payment Currency`	Categorical currency name/ code	Monetary interpretation requires currency-aware controls.
`Payment Format`	Categorical channel / payment instrument	Observed categories: **Cheque, Credit Card, ACH, Reinvestment, Wire, Bitcoin, Cash**. Validate taxonomy against source definitions.
`Is Laundering`	Binary target label (0/1)	Label provenance, adjudication criteria, and coverage are not documented. It must not be interpreted as a confirmed criminal finding without authoritative semantics.

4. STATISTICAL INFERENCE & HYPOTHESIS TESTING

All reported significance decisions use $\alpha = 0.05$. Statistical association is not proof of causation or criminal intent. These tests are exploratory; effect sizes, confidence intervals, diagnostics, multiple-testing controls, and independent validation should be added before relying on results for policy or customer-impacting decisions.

Test 1: Transaction amount vs. laundering status — Welch two-sample t-test

Question and scope: For records with `Receiving Currency == 'US Dollar'`, is mean `Amount Received` different between label 1 and label 0?

- **H_0:** $\mu L = \mu N$
- **H_1:** $\mu L \neq \mu N$
- **Selection rationale:** Welch's two-sample t-test does not require equal population variances and is more appropriate than the pooled-variance t-test when group variances/sample sizes may differ. Amounts are strongly right-skewed: the mean is far above the median. Welch's test does not make the raw amounts normally distributed; its inference relies on sampling behavior and can be sensitive to extreme values, so robust/bootstrap or transformed analyses are appropriate corroboration.
- **Notebook descriptive statistics:** Laundering-labelled mean **$5,461,686.87**, median **$3,286.21**; legitimate mean **$526,315.39**, median **$937.64**.
- **Finding:** Notebook decision: **reject H_0 ($p < 0.05$)**. The tested groups show statistically different average received amounts in this US Dollar subset. The large mean/median gaps indicate substantial right-skew and likely influence from extreme observations.
- **Compliance interpretation:** Amount is a potentially useful risk feature, not a stand-alone determination. Investigate currency normalization, extreme-value handling, sample duplication, target-label integrity, confidence intervals, and sensitivity to trimming/log-scale or robust methods. A significant difference does not imply that every large transaction is suspicious.

Test 2: Payment format dependency — chi-square test of independence

- **H_0:** Payment format and laundering status are independent.
- **H_1:** Payment format and laundering status are dependent.
- **Selection rationale:** A contingency-table association test is suitable for two categorical variables. The notebook uses a nominal-association chi-square procedure on the cross-tabulation of `Is Laundering` and `Payment Format`.
- **Finding:** Test statistic **21,130.71**; software printed p-value **0.0**. This is numerical underflow/rounding for an extremely small p-value, not a literal probability of zero. At $\alpha = 0.05$, reject H_0; the variables are statistically associated in the analyzed subset.
- **Compliance interpretation:** This supports testing payment format as a candidate feature. It does **not** by itself prove payment-channel preference causes laundering or establish "criminal laundering velocity"; velocity is a temporal rate and was not tested in this contingency analysis. Report observed/expected counts, standardized residuals, Cramér's V, and category-specific rates with uncertainty. Confirm expected cell counts and account for engineered prevalence, duplicates, and selection effects.

Test 3: Operational value across payment channels — Welch one-way ANOVA

- **H_0:** All payment-format population means for `Amount Received` are equal.
- **H_1:** At least one payment-format population mean differs.
- **Selection rationale:** There are more than two independent channel groups. Welch's one-way ANOVA (`use_var='unequal'`) accommodates unequal group variances, a material concern given highly skewed transaction amounts and heteroscedasticity. The ordinary OLS ANOVA is also shown in the notebook, but the reported conclusion below is based on Welch's test. No post-hoc pairwise procedure is supplied.
- **Finding:** Welch statistic **11.4434**, p-value **8.8958×10^{-12}**; reject H_0 at $\alpha = 0.05$. There is evidence that at least one payment-format mean differs. The OLS ANOVA also reports $F = 5.8200$ and $p = 0.000004$, but Welch's result is the unequal-variance analysis emphasized here.
- **Compliance interpretation:** The result indicates differences in transaction scale by channel in the analyzed data; it does not show that every channel differs from every other channel. Conduct

multiplicity-adjusted post-hoc comparisons, report group sizes/effect sizes and robust intervals, and normalize currency before comparing monetary values across currencies. Here analysis is on the US Dollar receiving subset due to the earlier filter.

Test 4: Systems drop verification — pre/post September 16, 2022

- **Data and windows:** Daily counts are computed from the **original** source timestamps, not from the resampled 150,000-row dataset. Pre-event window: September 1–15 (15 days); post-event window: September 16–26 (11 days). Welch's one-sided two-sample t-test is used, with unequal variances.
- **H_0:** $\mu pre \leq \mu post$
- **H_1:** $\mu pre > \mu post$
- **Finding:** Mean daily volume was **1,886,267.4** before the event and **268,860.7** after it. Welch t = **4.4625**, df = **22.37**, p = **9.41255×10^{-5}**. Reject H_0 at $\alpha = 0.05$; the observed drop is statistically significant under the test's assumptions.
- **Compliance interpretation:** This supports an operational data-quality/system-change investigation, not a claim that the mechanism or cause is established. A pre/post t-test alone does not prove a structural break: daily counts may be serially correlated, windows are short and unequal, and source coverage or system changes may explain the drop. Validate source completeness, ingestion and reporting pipelines, timezone/date handling, outages, and business-calendar effects. Use interrupted time-series/change-point methods and corroborating system logs before classifying this as a production incident.

5. SYSTEM CONCLUSION & FUTURE ROADMAP

Conclusion

The notebook demonstrates a workable exploratory pipeline and identifies candidate relationships between transaction amount, payment format, laundering labels, and daily system volume. These patterns, together with the scale of transaction monitoring and limitations of static rules, provide a rationale to investigate a supervised ML classifier for risk ranking and alert prioritization. They are not sufficient evidence of predictive performance or operational fitness: no trained classifier, holdout evaluation, leakage assessment, calibration, alert-capacity analysis, or investigator validation is present in the notebook.

Before model deployment, establish label governance and representativeness; prevent entity/time and oversampling leakage; evaluate temporal out-of-sample performance; measure precision/recall and PR-AUC at realistic prevalence; assess calibration, stability, subgroup outcomes, and false-negative/false-positive impacts; set risk-based thresholds with compliance ownership; and maintain monitoring, change control, and independent validation. Model alerts should be explainable, reviewable, and subject to documented human decisions.

Future roadmap

1. **Real-time streaming integration:** Ingest transactions through governed streaming interfaces, apply schema and quality checks, manage latency and replay/idempotency, and monitor missingness, drift, and service availability. Retain reproducible feature lineage and auditable alert records.
2. **Network graph analytics:** Link accounts, banks, devices, counterparties, and transactions to identify shared entities, circular flows, rapid pass-through, and multi-hop relationships. Apply privacy, entity-resolution quality, temporal context, and investigator-verifiable graph explanations.
3. **Explainable AI (XAI) dashboards:** Present leading contributing factors, relevant transaction/counterparty context, data-quality indicators, uncertainty, and model/version details to compliance auditors and investigators. Support feedback and disposition capture while preventing explanations from being treated as proof of wrongdoing.

4. **Governance and validation:** Define model inventory and ownership, independent validation, documented limitations, threshold approval, ongoing drift/performance monitoring, controlled retraining, audit trails, access controls, and human escalation procedures consistent with applicable institutional policy and regulation.

Audit note: Reported numerical findings are transcribed from notebook outputs. The notebook does not provide confidence intervals, effect-size estimates, post-hoc tests, formal assumption diagnostics, independent replication, data lineage beyond the named files, or a complete model-risk package. Those gaps should be closed before any operational or regulatory reliance.